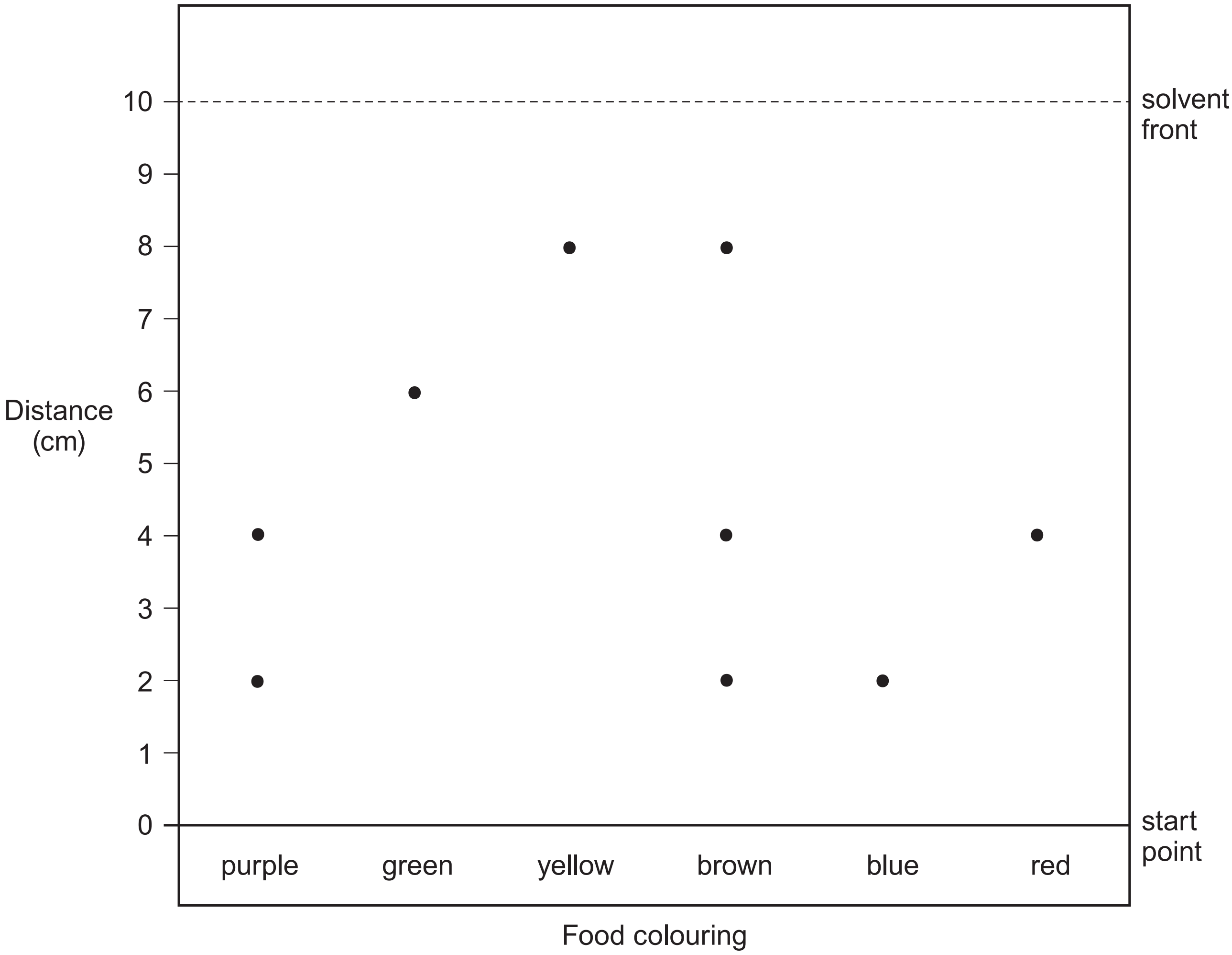
Examiner only

2. Analytical scientists work in a wide range of different industries and agencies.

(a) The Food Standards Agency (FSA) funded a project to analyse the colours used in food and drinks.

The diagram below shows a chromatogram of the dyes present in different food colourings.



(i) It was thought that the dye in the blue colouring would not be present in other colourings.
Explain whether you agree. [2]

....

....

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(ii) Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

to calculate the R_f value for the dye in the **red** colouring. [3]

R_f = ...

(b) Analytical scientists also identify unknown compounds.
There were three bottles, each containing a solution of one of the following compounds.

copper(II) sulfate	lithium carbonate	sodium chloride

(i) State which of these compounds would produce a yellow flame in a flame test. [1]

....

(ii) I. State which of these compounds would produce bubbles of gas when added to hydrochloric acid. [1]

....

II. The gas turns limewater milky. Name this gas. [1]

....

(iii) 1 cm^3 of silver nitrate solution is added to three different test tubes, each containing a solution of one of the compounds in the box above.

I. State which of these compounds will give a white precipitate. [1]

....

II. Complete the risk assessment below for silver nitrate. [2]

Hazard	Risk	Control measure
silver nitrate is irritant	could cause eye damage if splashes in the eyes when	....

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